

Practical 8 - Time-dependent variables

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `jas`, `jas1`

R packages required: `survival`, `tidyverse`

```
library(survival)
library(tidyverse)
```

Introduction

In this practical we will perform analyses of data from the Stanford Heart Transplant programme. We saw examples from this in the lecture. The data records information on individuals in a programme in which they were waiting for a heart transplant. It includes 103 individuals who were followed up from the date of their acceptance onto the programme to death or censoring. A version of this classic data set is available in the `survival` package in R. We will use the versions called `jas` and `jas1`, where the name refers to the *Journal of the American Statistical Association* where results from analysis of the data were published (e.g. Crowley and Hu (1977)).

Variables we will use in the `jas` data (one row per individual) are:

Variable	Description
<code>futime</code>	Time in days to death or censoring.
<code>fustat</code>	Status at the end of follow-up. Death: 1, Censoring: 0
<code>transplant</code>	Indicator of whether the person ever received a transplant
<code>wait.time</code>	Time in days to transplant
<code>age</code>	Age in years at joining the programme
<code>surgery</code>	Prior bypass surgery (No: 0, Yes: 1)

Variable	Description
<code>tx.date</code>	Transplant date

Variables we will use in the `jasal` data (more than one row per individual) are:

Variable	Description
<code>id</code>	Patient id
<code>start</code>	Start time for this time interval
<code>stop</code>	Stop time for this time interval
<code>event</code>	Status at the end of the interval. Death: 1, Censoring: 0

Aims

The aims of this session are:

- Use data sets with one row per individual and multiple rows per individual for analysis involving time-dependent variables.
- Understand how immortal time bias can arise.
- Fit and interpret Cox models including time-dependent explanatory variables.
- Use the landmarking approach for analysis of time-dependent variables.

Question 1

We will begin by investigating what is in the data and how the data are arranged.

- Load the datasets `jasal` and `jasal1`. How many records (rows) are there in the two datasets? And how many individuals?
- How many individuals received a transplant?
- Look at the data for individuals 6-10 in the two data sets and see how the variables `wait.time`, `futime`, `transplant` in `jasal` are connected to the variables `start`, `stop` and `transplant` in `jasal1`.

SOLUTION

```
dim(jasal) #n=103 individuals, 103 rows of data
```

```
[1] 103 14
```

```
dim(jasa1) #170 rows
```

```
[1] 170    8
```

```
length(unique(jasa1$id))#n=103 individuals
```

```
[1] 103
```

```
table(jasa$transplant)#69 transplants
```

```
 0  1  
34 69
```

```
table(jasa1$transplant)
```

```
 0  1  
101 69
```

```
#Look individuals 6-10
```

```
jasa[6:10,c("accept.dt","tx.date","fu.date","fustat",  
            "wait.time","futime","transplant")]
```

	accept.dt	tx.date	fu.date	fustat	wait.time	futime	transplant
6	1968-06-13	<NA>	1968-06-15	1	NA	2	0
7	1968-07-12	1968-08-31	1970-05-17	1	50	674	1
8	1968-08-01	<NA>	1968-09-09	1	NA	39	0
9	1968-08-09	<NA>	1968-11-01	1	NA	84	0
10	1968-08-11	1968-08-22	1968-10-07	1	11	57	1

```
#Same 5 individuals in the long format data
```

```
jasa1[jasa1$id%in%c(6:10),c("id","start","stop","event","transplant")]
```

	id	start	stop	event	transplant
5	6	0	2	1	0
6	7	0	50	0	0

104	7	50	674	1	1
7	8	0	39	1	0
8	9	0	84	1	0
9	10	0	11	0	0
105	10	11	57	1	1

- There are 103 people in the data, and 69 people had a heart transplant.
- The dataset `java1` has 170 rows because people who have a transplant have two rows of data - one row for their pre-transplant follow-up and one row for their post-transplant follow-up.

Question 2

- Using the data with one row per individual (`java`) perform a Kaplan-Meier analysis by transplant status, and perform a Cox regression to estimate the hazard ratio for transplant.
- What is wrong with these analyses? How long are individuals 7 and 10 ‘immortal’ in the way that the data are being used here?

SOLUTION

```
#naive Cox regression with immortal time bias
#treating transplant status as though known at time zero
cox.naive<-coxph(Surv(futime, fustat)~transplant,data=java)
summary(cox.naive)
```

Call:

```
coxph(formula = Surv(futime, fustat) ~ transplant, data = java)
```

n= 103, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
transplant	-1.3234	0.2662	0.2438	-5.428	5.69e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
transplant	0.2662	3.756	0.1651	0.4293

Concordance= 0.668 (se = 0.025)

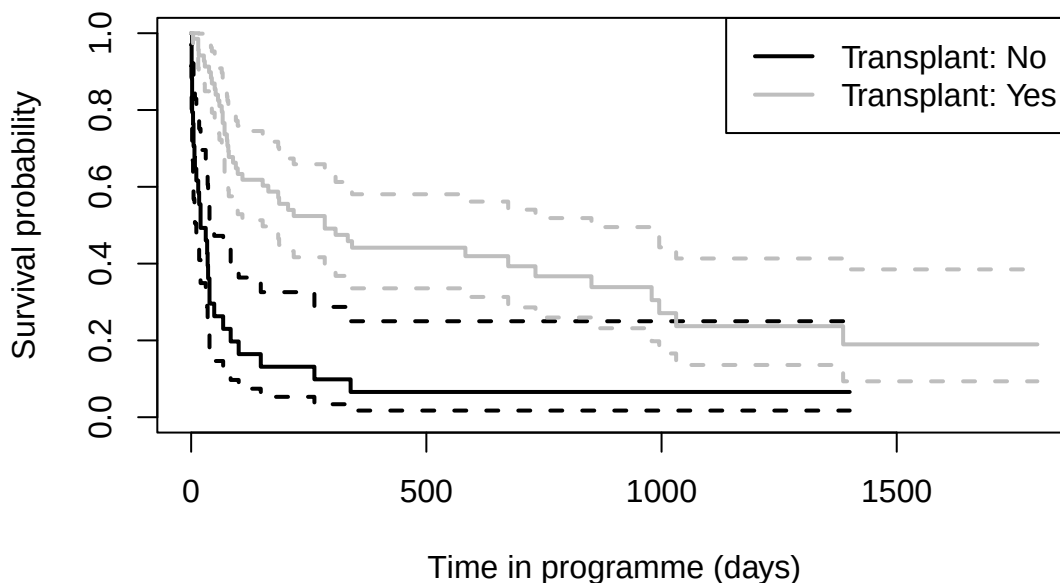
Likelihood ratio test= 25.95 on 1 df, p=4e-07

Wald test = 29.47 on 1 df, p=6e-08

Score (logrank) test = 33.38 on 1 df, p=8e-09

```
#Naive Kaplan-Meier analysis,
#treating transplant status as though known at time zero
km.naive<-survfit(Surv(futime, fustat)~transplant,data=jasa)

#Plot
plot(km.naive,
     xlab="Time in programme (days)",ylab="Survival probability",
     col=c("black","grey"),lwd=2,conf.int=T,)
legend(x="topright",c("Transplant: No","Transplant: Yes"),
     col=c("black","grey"),lty=1,lwd=2)
```



```
#Take a look at individuals 7 and 10
jasa[c(7,10),c("futime","fustat","transplant","wait.time")]
```

	futime	fustat	transplant	wait.time
7	674	1	1	50
10	57	1	1	11

- The Kaplan-Meier plot suggests better survival for transplanted individuals, as does the HR estimate of 0.27 for transplant versus no transplant. There is a strong association between transplant status and the hazard, with the HR having 95% CI 0.17 to 0.43. However, this analysis ignores the time-dependent nature of transplant status, and the results are therefore subject to immortal time bias.
- Person 7 had a transplant at day 50 and person 10 at day 11. In the above analysis they were considered as transplanted from time 0, hence they were 'immortal' up until their time of transplant.

Question 3

- We will now consider transplant as the time-dependent variable that it is. Write down the form of a Cox proportional hazards model for death, with transplant status as a time-dependent variable.
- Fit the model using the data in the long format (`jasal`) and interpret the results. Think about what the format of the `Surv()` function should be to account for the long format of the data.

SOLUTION

```
#Cox regression using time-updated transplant status
cox.td<-coxph(Surv(start, stop, event)~transplant,data=jasal)
summary(cox.td)
```

Call:

```
coxph(formula = Surv(start, stop, event) ~ transplant, data = jasal)
```

n= 170, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
transplant	0.1252	1.1333	0.3008	0.416	0.677

	exp(coef)	exp(-coef)	lower .95	upper .95
transplant	1.133	0.8824	0.6285	2.044

Concordance= 0.511 (se = 0.025)

Likelihood ratio test= 0.17 on 1 df, p=0.7

Wald test = 0.17 on 1 df, p=0.7

Score (logrank) test = 0.17 on 1 df, p=0.7

- The form of the model is $h(t|X(t)) = h_0(t) \exp(\beta X(t))$, where $X(t)$ denotes transplant status at time t , taking value 1 if a person received a transplant before time t and value 0 otherwise.
- The HR for transplant is 1.13. The interpretation is that having received a transplant prior to time t is associated with a 13% higher hazard of death at time t compared with not having received a transplant before time t . There is no evidence against the null hypothesis of no association between current transplant status and the hazard, as the p-value is large.

Question 4

In this question we will extend the model in question 3 to allow the hazard to depend not only on current transplant status, but also on the time since transplant. Use the following steps:

- Start by splitting the long form data (`jasal`) using the unique event times as the cut points. You can do this using `survsplit`, which we have used in earlier practicals.
- Generate a time-dependent variable that represents the time since transplant in each row, setting this to 0 prior to transplant.
- Categorise the time since transplant into three periods: period 1: 0-90 days, period 2: 91-186 days, period 3: >186 days
- Fit a Cox model using time-dependent transplant status plus its interaction with the time period.
- Discuss how the association between transplant and the hazard depends on time-since-transplant.

SOLUTION

```
#Generate time since transplant
#Need to split the data further to do this
event.times<-sort(unique(jasal$stop[jasal$event==1]))
jasal.split<-survSplit(Surv(start, stop, event)~.,data=jasal,cut=event.times)

jasal.split<-jasal.split%>%
  mutate(time.since.tx=ifelse(transplant==1,stop-start,0))
jasal.split<-jasal.split%>%group_by(id)%>%
  mutate(time.since.tx=cumsum(time.since.tx))

#Group time since transplant into 3 periods:
#p1: up to 90 days, p2: 91 days to 186 months; p3: >186 days
jasal.split$time.since.tx.grp<-cut(jasal.split$time.since.tx,
                                  breaks = c(0,90,186,10000),
                                  include.lowest = T)

jasal.split$transplant.p1<-jasal.split$transplant*
  I(jasal.split$time.since.tx<=90)
jasal.split$transplant.p2<-jasal.split$transplant*
  I(jasal.split$time.since.tx>90 & jasal.split$time.since.tx<=186)
jasal.split$transplant.p3<-jasal.split$transplant*
  I(jasal.split$time.since.tx>186)

#Cox regression using time-updated transplant status, allowing the
#impact of transplant on the hazard to vary with time since transplant.
```

```
#This can be done in two equivalent ways

#cox.td2: helps us look at evidence for interaction
#between transplant and time since transplant
cox.td2<-coxph(Surv(start, stop, event)~
               transplant+transplant:time.since.tx.grp,
               data=jasal.split)
summary(cox.td2)
```

Call:

```
coxph(formula = Surv(start, stop, event) ~ transplant + transplant:time.since.tx.grp,
      data = jasal.split)
```

n= 3606, number of events= 75

	coef	exp(coef)	se(coef)	z
transplant	0.3200	1.3771	0.3185	1.005
transplant:time.since.tx.grp(90,186]	-1.0786	0.3401	0.7412	-1.455
transplant:time.since.tx.grp(186,1e+04]	-0.8220	0.4395	0.6888	-1.193

Pr(>|z|)

transplant	0.315
transplant:time.since.tx.grp(90,186]	0.146
transplant:time.since.tx.grp(186,1e+04]	0.233

	exp(coef)	exp(-coef)	lower .95
transplant	1.3771	0.7261	0.73767
transplant:time.since.tx.grp(90,186]	0.3401	2.9405	0.07956
transplant:time.since.tx.grp(186,1e+04]	0.4395	2.2751	0.11395

upper .95

transplant	2.571
transplant:time.since.tx.grp(90,186]	1.454
transplant:time.since.tx.grp(186,1e+04]	1.695

Concordance= 0.528 (se = 0.027)

Likelihood ratio test= 2.93 on 3 df, p=0.4

Wald test = 3.02 on 3 df, p=0.4

Score (logrank) test = 3.05 on 3 df, p=0.4

```
#cox.td3: helps us see the HRs for transplant in the three time periods
cox.td3<-coxph(Surv(start, stop, event)~
               transplant.p1+transplant.p2+transplant.p3,
```

```
data=jasa1.split)
summary(cox.td3)
```

Call:

```
coxph(formula = Surv(start, stop, event) ~ transplant.p1 + transplant.p2 +
      transplant.p3, data = jasa1.split)
```

n= 3606, number of events= 75

	coef	exp(coef)	se(coef)	z	Pr(> z)
transplant.p1	0.3200	1.3771	0.3185	1.005	0.315
transplant.p2	-0.7586	0.4683	0.7252	-1.046	0.296
transplant.p3	-0.5020	0.6053	0.6360	-0.789	0.430

	exp(coef)	exp(-coef)	lower .95	upper .95
transplant.p1	1.3771	0.7261	0.7377	2.571
transplant.p2	0.4683	2.1352	0.1130	1.940
transplant.p3	0.6053	1.6521	0.1740	2.106

Concordance= 0.528 (se = 0.027)

Likelihood ratio test= 2.93 on 3 df, p=0.4

Wald test = 3.02 on 3 df, p=0.4

Score (logrank) test = 3.05 on 3 df, p=0.4

- The model fitted (cox.td2) is of the form:

$$h(t|X(t)) = h_0(t) \exp\{\beta_1 X(t) + \beta_2 X(t)I(90 < Z(t) \leq 186) + \beta_3 X(t)I(Z(t) > 186)\}$$

or equivalently (cox.td3)

$$h(t|X(t)) = h_0(t) \exp\{\beta_1^* X(t)I(Z(t) \leq 90) + \beta_2^* X(t)I(90 < Z(t) \leq 186) + \beta_3^* X(t)I(Z(t) > 186)\}$$

where $X(t)$ denotes transplant status at time t , $Z(t)$ denotes time since transplant (in days) at time t , and $I(Z(t) \leq 90)$ (for example) is the binary indicator of whether $Z(t) \leq 90$.

- This results in the following point estimates from cox.td2: $e^{\hat{\beta}_1} = 1.38$, $e^{\hat{\beta}_2} = 0.34$, $e^{\hat{\beta}_3} = 0.43$. The point estimates from cox.td3 are $e^{\hat{\beta}_1^*} = 1.38$, $e^{\hat{\beta}_2^*} = 0.47$, $e^{\hat{\beta}_3^*} = 0.61$, and these are the HRs for transplant in each period post-transplant versus no transplant.
- The point estimates indicate that within the first 90 days post-transplant, transplanted individuals had a 37% higher hazard of mortality compared with untransplanted individuals. In the period 91-186 days post-transplant, transplanted individuals are estimated to have a 53% lower hazard compared with being untransplanted. In the period >186 days post-transplant the hazard of mortality for transplanted individuals was estimated

to be 39% lower than that of untransplanted individuals. The confidence intervals all include 1, indicating no evidence of an association between transplant status and the hazard in any of the three periods post-transplant.

Question 5

In this question we will apply the landmarking approach, using a landmark time of 90 days. Go through the following steps:

- Using the `java` dataset, restrict the data to people who remain alive 90 days after joining the programme.
- Reset the time origin to be day 90. This means that you need to modify the follow-up time variable `futime` to reflect the change in the time origin.
- Generate a variable that indicates whether or not a person received a transplant before day 90, i.e. their transplant status at the landmark time.
- Generate Kaplan-Meier survival curves by transplant status at the landmark time based on this modified data, and fit a Cox model.
- What happens if you change the landmark time to 30 days?

SOLUTION

```
#landmark time: 90 days after joining the programme
lm<-90

#drop people who have the event before 90 days
java.lm<-java
java.lm<-java.lm[java.lm$futime>lm,]

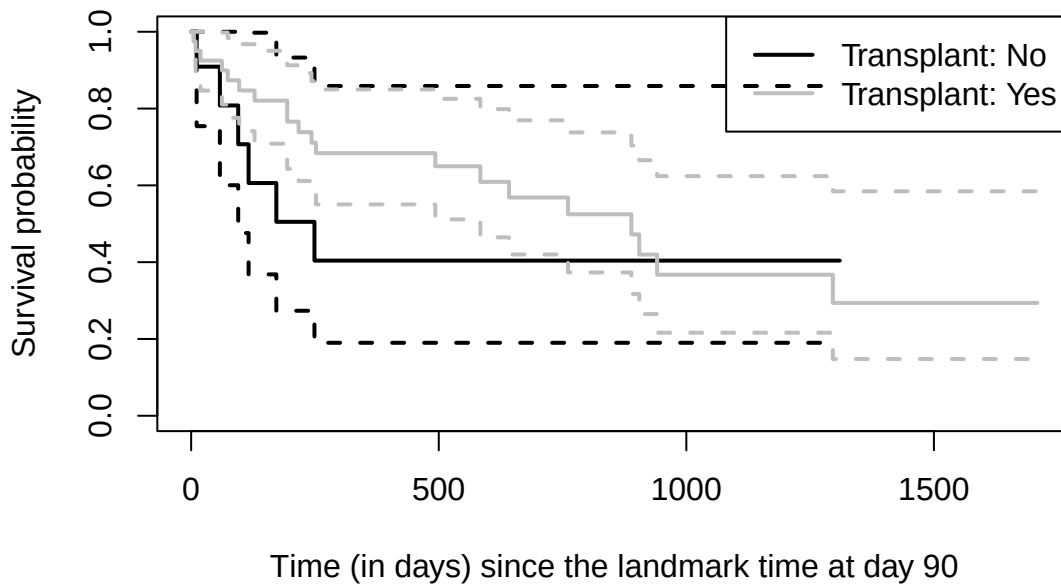
#follow-up time relative to the landmark time
java.lm$futime.lm<-java.lm$futime-lm

#transplant status at the landmark time (0 or 1)
java.lm$transplant.lm<-ifelse(
  java.lm$transplant==1 & java.lm$wait.time>lm,0,
  java.lm$transplant)

#Kaplan-Meier using transplant status at the landmark time
km.lm<-survfit(Surv(futime.lm, fustat)~transplant.lm,data=java.lm)

plot(km.lm,
      xlab="Time (in days) since the landmark time at day 90",
      ylab="Survival probability",
```

```
col=c("black","grey"),lwd=2,conf.int=T,)
legend(x="topright",c("Transplant: No","Transplant: Yes"),
      col=c("black","grey"),lty=1,lwd=2)
```



```
#Cox regression using transplant status at the landmark time
cox.lm<-coxph(Surv(futime.lm, fustat)~transplant.lm,data=jasa.lm)
summary(cox.lm)
```

Call:

```
coxph(formula = Surv(futime.lm, fustat) ~ transplant.lm, data = jasa.lm)
```

n= 51, number of events= 26

	coef	exp(coef)	se(coef)	z	Pr(> z)
transplant.lm	-0.5567	0.5731	0.4728	-1.178	0.239

	exp(coef)	exp(-coef)	lower .95	upper .95
transplant.lm	0.5731	1.745	0.2269	1.448

Concordance= 0.566 (se = 0.047)

Likelihood ratio test= 1.26 on 1 df, p=0.3

Wald test = 1.39 on 1 df, p=0.2

Score (logrank) test = 1.42 on 1 df, p=0.2

- Among people who survived to day 90, those who received a transplant prior to day 90 had a 43% lower hazard compared with those who had not received a transplant prior to day 90, though the confidence interval is wide and includes 1.
- The focus here was on transplant status at day 90. Some of the people who had not received a transplant prior to day 90 received one during their subsequent follow-up, and that is part of what contributes to their survival after day 90.
- The HR with landmark time at day 90 was 0.57 (95% CI 0.23 to 1.45). Changing the landmark time to day 30 gives a HR of 0.92 (95% CI 0.53 to 1.59).

Question 6

Suppose we are interested in studying post-transplant survival, and how the following factors are associated with post-transplant survival: age (**age**), whether or not the person had had prior bypass surgery (**surgery**), calendar year in which the transplant was received (using **tx.date**).

- Starting from the **java** data, set-up a new dataset that enables you to study post-transplant survival.
- Obtain a Kaplan-Meier plot for post-transplant survival. What are the estimated probabilities of survival to 30 days and 1 year post-transplant?
- Fit a Cox model including **age**, **surgery** and calendar year of transplant. Interpret the results.
- Investigate the assumptions of the model using Schoenfeld residuals and Martingale residuals.

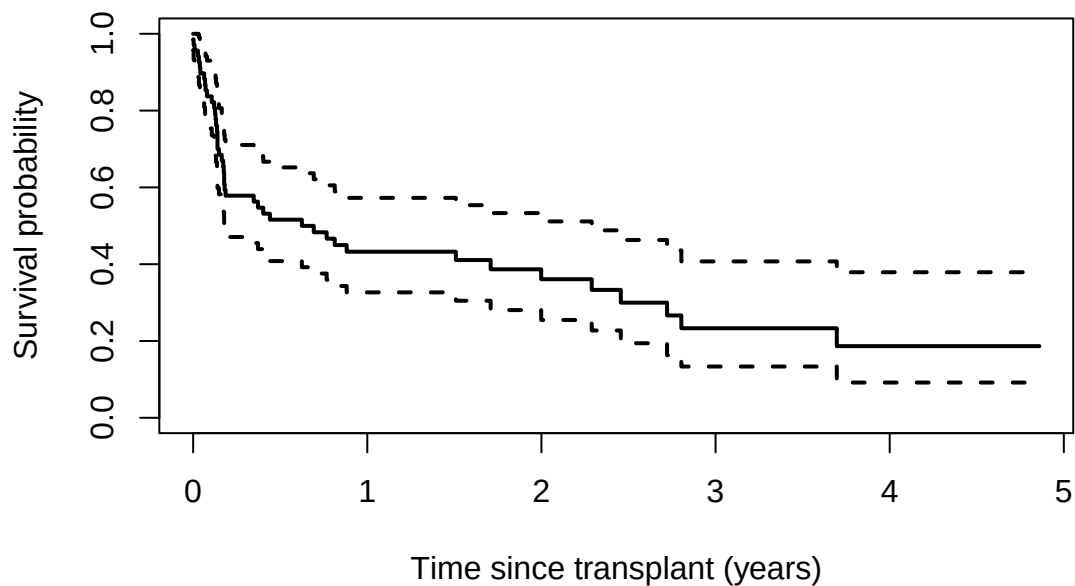
SOLUTION

```
#Restrict to people who have a transplant
java.tx<-java[java$transplant==1,]

#Generate follow-up time relative to the time of having a transplant
java.tx$time_since_tx<-java.tx$futime-java.tx$wait.time

#Kaplan-Meier plot and estimates of probabilities
km.tx<-survfit(Surv(time_since_tx/365.25,fustat)~1,data=java.tx)

plot(km.tx,
      xlab="Time since transplant (years)",ylab="Survival probability",
      lwd=2,conf.int=T)
```



```
summary(km.tx, times=c(30/365.25, 1))
```

Call: `survfit(formula = Surv(time_since_tx/365.25, fustat) ~ 1, data = jasa.tx)`

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
0.0821	56	11	0.837	0.0449	0.754	0.930
1.0000	24	26	0.432	0.0620	0.327	0.573

```
#Generate year of transplant
jasa.tx$year_tx<-as.numeric(format(jasa.tx$tx.date,"%Y"))

#Cox model
mod.tx<-coxph(Surv(time_since_tx,fustat)~age+surgery+year_tx,data=jasa.tx)
summary(mod.tx)
```

Call:

```
coxph(formula = Surv(time_since_tx, fustat) ~ age + surgery +
      year_tx, data = jasa.tx)
```

n= 69, number of events= 45

	coef	exp(coef)	se(coef)	z	Pr(> z)
age	0.04726	1.04839	0.02168	2.179	0.0293 *
surgery	-0.85121	0.42690	0.45027	-1.890	0.0587 .

```
year_tx -0.03916    0.96160  0.09131 -0.429    0.6680
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0484	0.9538	1.0048	1.094
surgery	0.4269	2.3425	0.1766	1.032
year_tx	0.9616	1.0399	0.8040	1.150

```
Concordance= 0.644 (se = 0.046 )
```

```
Likelihood ratio test= 11.62 on 3 df,  p=0.009
```

```
Wald test = 10.49 on 3 df,  p=0.01
```

```
Score (logrank) test = 11 on 3 df,  p=0.01
```

```
#Schoenfeld residuals
```

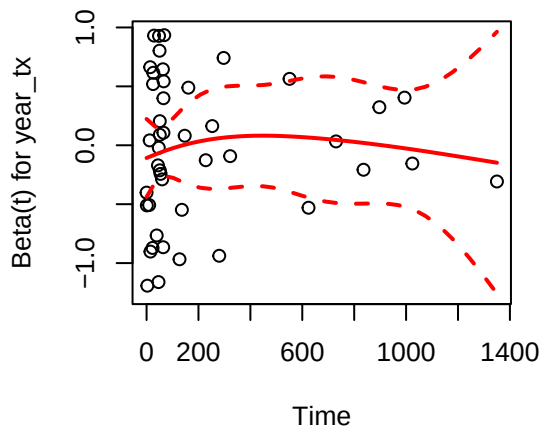
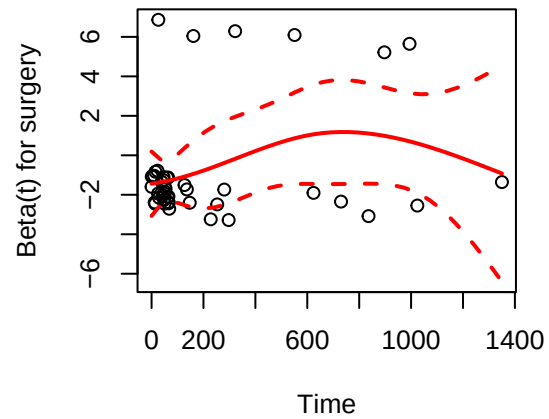
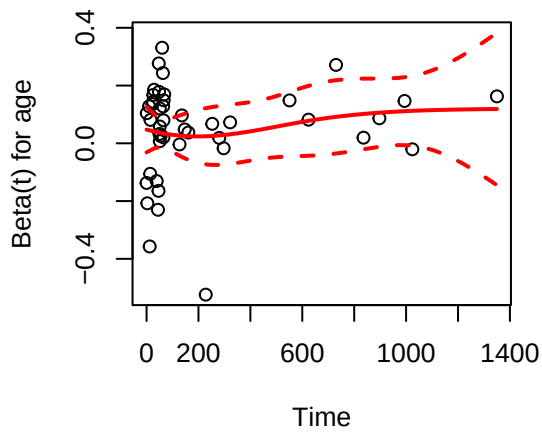
```
sch.resid<-cox.zph(mod.tx, transform = 'identity')
```

```
sch.resid
```

	chisq	df	p
age	0.891	1	0.35
surgery	1.806	1	0.18
year_tx	0.088	1	0.77
GLOBAL	2.857	3	0.41

```
par(mfrow=c(2,2))
```

```
plot(sch.resid,col="red",lwd=2)
```

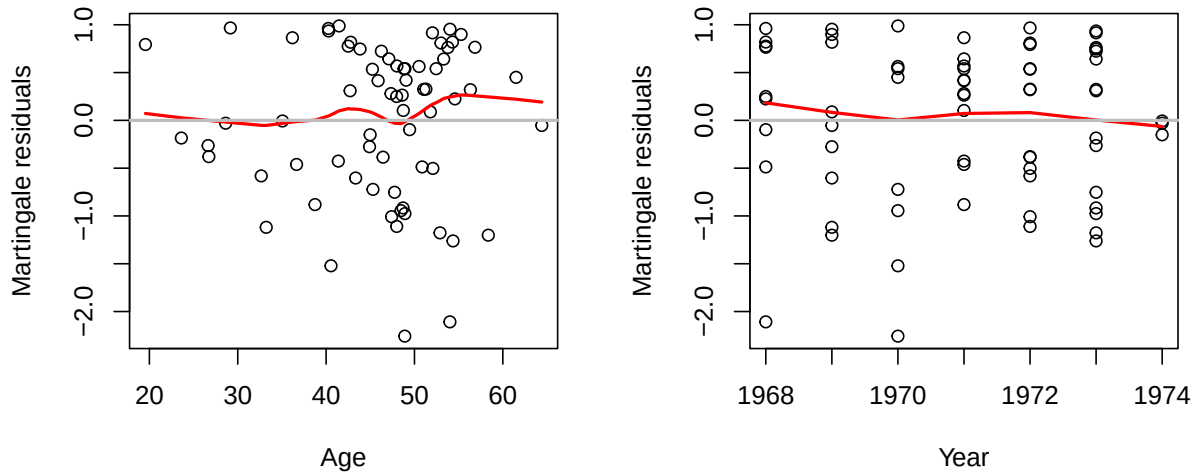


```
#Martingale residual plots for age and year
mgale.res<-resid(mod.tx,type="martingale")

par(mfrow=c(1,2))

plot(jasa.tx$age,mgale.res,xlab="Age",ylab="Martingale residuals")
lines(lowess(jasa.tx$age,mgale.res),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")

plot(jasa.tx$year,mgale.res,xlab="Year",ylab="Martingale residuals")
lines(lowess(jasa.tx$year,mgale.res),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")
```



- To study post-transplant survival, we restrict to people who received a transplant. The time origin is set to be their time of transplant.
- The estimated probabilities of survival to 1 month post-transplant and 1 year post transplant are respectively 84% and 43%.
- In the multivariable Cox model, a 1 year increase in age is associated with a 4.8% (95% CI 0.5% to 9.4%) increase in the hazard of death post transplant, conditional on surgery and year. Conditional on age and year, having had a prior bypass surgery is associated with a 57% reduction in the hazard, and this is of borderline significance (HR 0.43, 95% CI 0.18 to 1.03). Conditional on age and surgery, the year of receiving the transplant (over this period from 1968 to 1974) was not associated with post-transplant survival (HR 0.96 for a 1 year increase, 95% CI 0.80 to 1.15).
- Based on the Schoenfeld plots and corresponding tests there is no evidence against the proportional hazards assumption for age, surgery or year. Looking at the Martingale residual plots for age and year, there is little suggestion that including age and year in a linear form in the model was not appropriate.